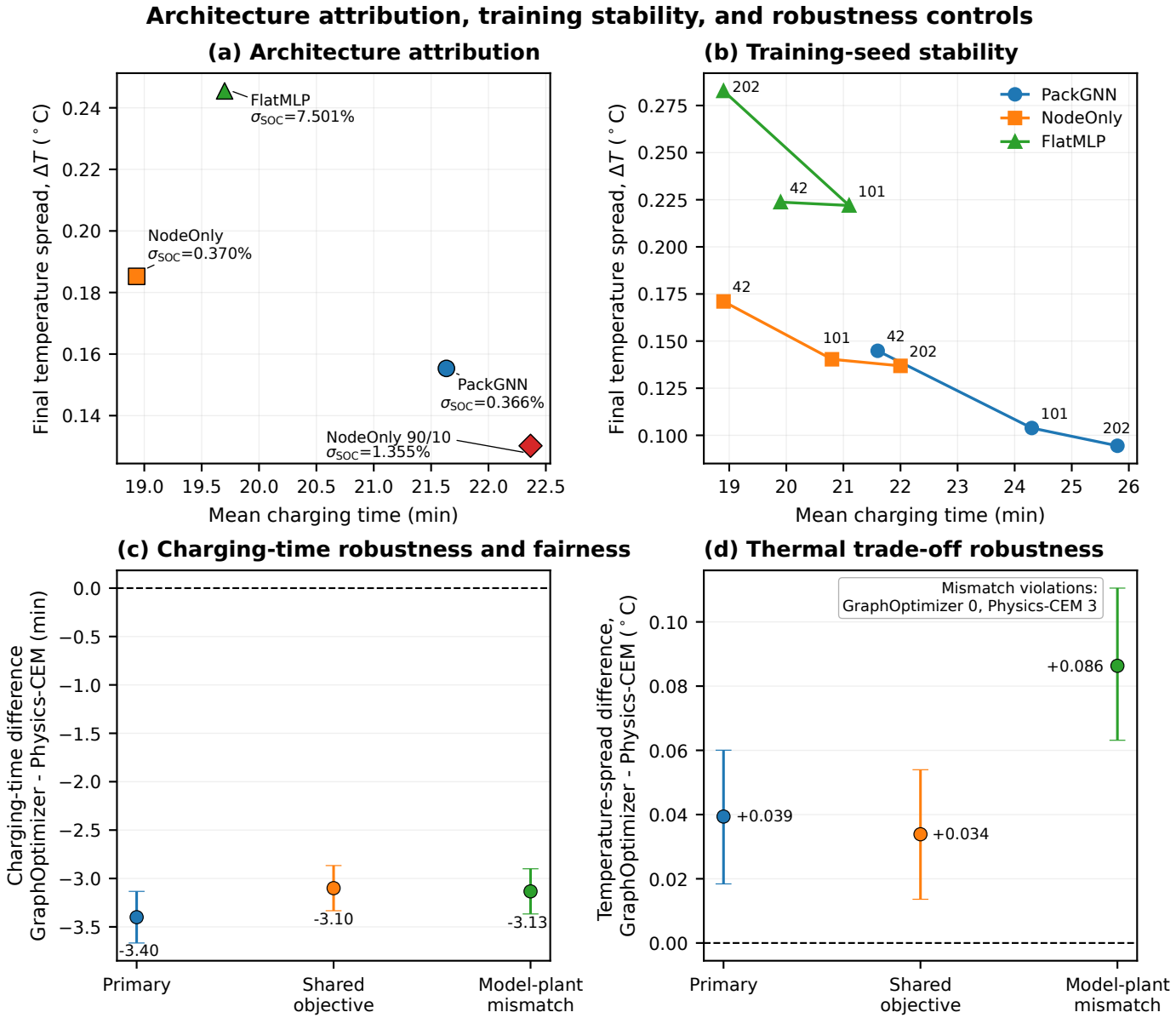

Panels (c,d) report paired GraphOptimizer-minus-Physics-CEM mean differences with 95% paired-bootstrap confidence intervals (20,000 resamples). Negative charging-time differences favour GraphOptimizer; positive thermal differences indicate larger final inter-cell temperature spread.